

● EASY

● ADVANCED MATH

Advanced Math

01

10 questions · ~15 min

SAT QUESTION BANK

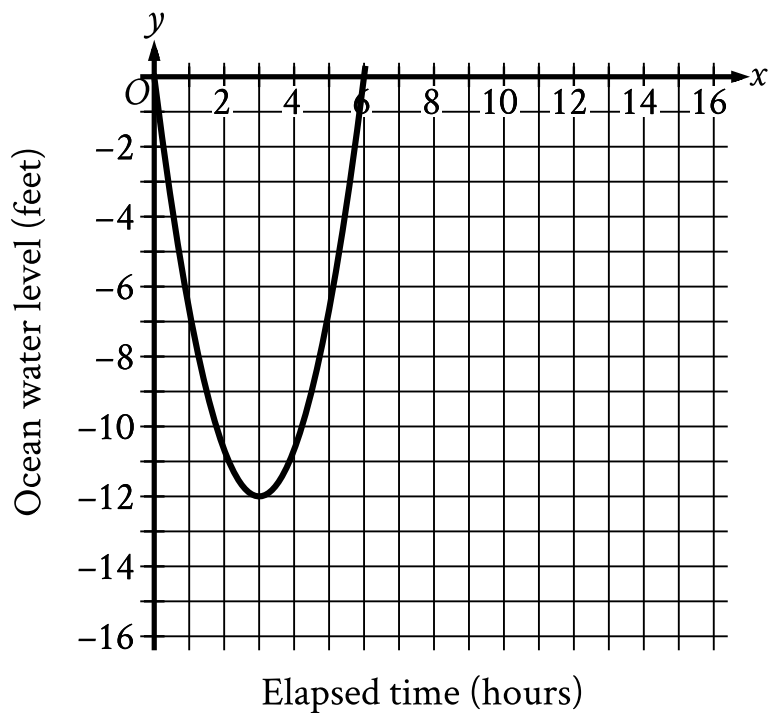
2026-05-29

Q1

ADVANCED MATH

Which expression is equivalent to $5x^5 - 6x^4 + 8x^3$?

- A. $x^4 5x - 6$
- B. $x^3 5x^2 - 6x + 8$
- C. $8x^3 5x^2 - 6x + 1$
- D. $6x^5 - 6x^4 + 8x^3 + 1$



- The parabola opens upward.
- The parabola passes through the following points:
 - (0 comma 0)
 - (3 comma negative 12)
 - (6 comma 0)

Scientists recorded data about the ocean water levels at a certain location over a period of 6 hours. The graph shown models the data, where $y = 0$ represents sea level. Which table gives values of x and their corresponding values of y based on the model?

A.

x	y
0	-12
0	3
3	6

B.

x	y
0	0
3	12
0	-6

C.

x	y
0	0
3	-12
6	0

D.

x	y
0	0
12	3
-6	0

$$y - 57 = px$$

The given equation relates the positive numbers p , x , and y . Which equation correctly expresses y in terms of p and x ?

A. $y = 57x + p$

B. $y = px + 57$

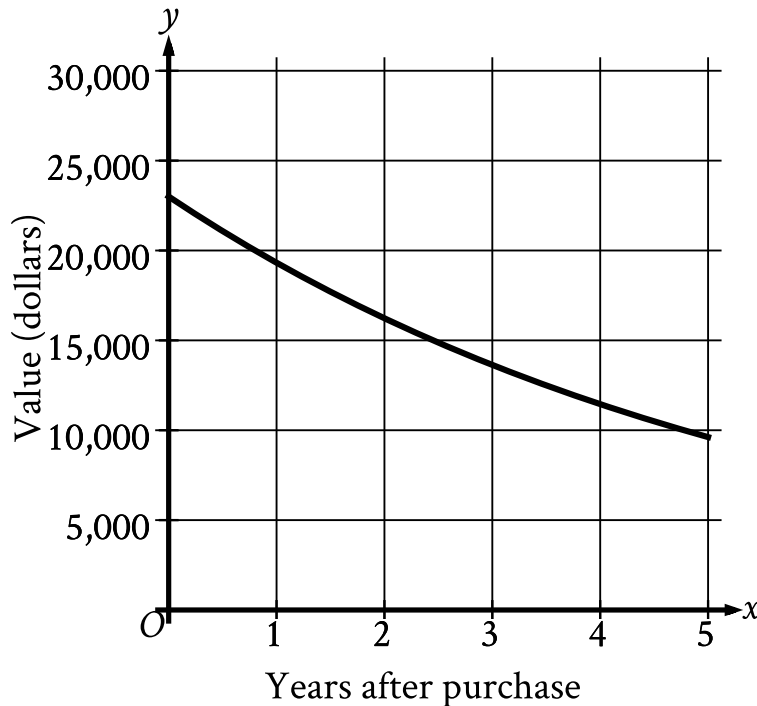
C. $y = 57px$

D. $y = \frac{px}{57}$

Which expression is a factor of $2x^2 + 38x + 10$?

- A. 2
- B. $5x$
- C. $38x$
- D. $2x^2$

The graph shows the predicted value y , in dollars, of a certain sport utility vehicle x years after it is first purchased.



- The curve is in quadrant 1.
- The curve trends down gradually from left to right.
- The curve passes through the following approximate points:
 - (0 comma 23,000)
 - (1 comma 19,320)
 - (3 comma 13,632)
 - (5 comma 9,619)

Which of the following is closest to the predicted value of the sport utility vehicle 3 years after it is first purchased?

- A. \$ 9,619
- B. \$ 13,632

C. \$ 19,320

D. \$ 23,000

$$y = 5x + 4$$

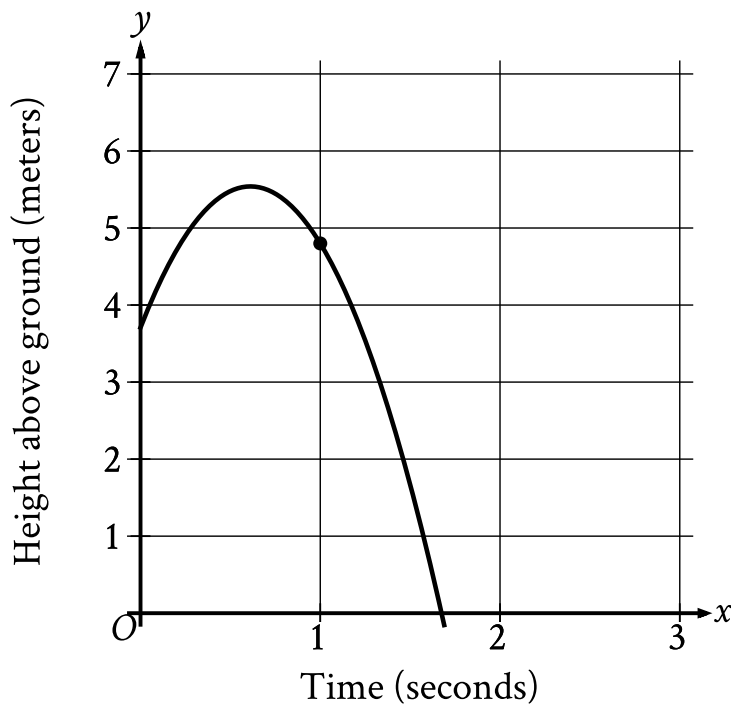
$$y = 5x^2 + 4$$

Which ordered pair x, y is a solution to the given system of equations?

- A. 0,0
- B. 0,4
- C. 8,44
- D. 8,84

Which expression is equivalent to $34x + 34y$?

- A. $34xy$
- B. $34(x + y)$
- C. $68y$
- D. $68x$



- All values are approximate.
- The parabola opens downward.
- The vertex of the parabola is at (0.6 comma 5.5).
- The parabola passes through the following points:
 - (0.0 comma 3.8)
 - (0.6 comma 5.5)
 - (1.0 comma 4.8)
 - (1.7 comma 0.0)

The graph shows the height above ground, in meters, of a ball x seconds after the ball was launched upward from a platform. Which statement is the best interpretation of the marked point 1.0, 4.8 in this context?

- A. 1.0 second after being launched, the ball's height above ground is 4.8 meters.
- B. 4.8 seconds after being launched, the ball's height above ground is 1.0 meter.

- C. The ball was launched from an initial height of 1.0 meter with an initial velocity of 4.8 meters per second.
- D. The ball was launched from an initial height of 4.8 meters with an initial velocity of 1.0 meter per second.

$$x + y = 12$$

$$y = x^2$$

If (x, y) is a solution to the system of equations above, which of the following is a possible value of x ?

A. 0

B. 1

C. 2

D. 3

Which expression is equivalent to $x^2 + 3x - 40$?

A. $x - 4x + 10$

B. $x - 5x + 8$

C. $x - 8x + 5$

D. $x - 10x + 4$